17. Networking Basics: IP, DNS, Ports

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Phase: Phase 6 — Networking
Estimated time: 1–2 days
Slug: `17-networking-basics`

Learning objectives

- Explain IP, CIDR, ports at a practical level
- Query DNS with dig/nslookup/getent
- Inspect addresses with ip
- Test connectivity with ping/curl

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

Essentials

- **IP address**: host identity on a network
- **Port**: endpoint for a service (22 ssh, 80 http, 443 https)
- **DNS**: names → addresses

```
ip addr
ip route
dig example.com +short
getent hosts example.com
ping -c 3 1.1.1.1
curl -I https://example.com
```

On older systems you may see `ifconfig`/`route` — prefer `ip` on modern Linux.

Common mistakes

- Assuming ping failure means everything is down (ICMP may be blocked)
- Hardcoding IPs without DNS strategy
- Opening all ports “temporarily”

Exercises

Work in `exercises/17-networking-basics/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/17-networking-basics/` until you have an honest failing attempt.

Self-check

1. What is a port?
2. How resolve DNS from CLI?
3. ip addr shows what?

Further reading

- https://man7.org/linux/man-pages/man8/ip.8.html